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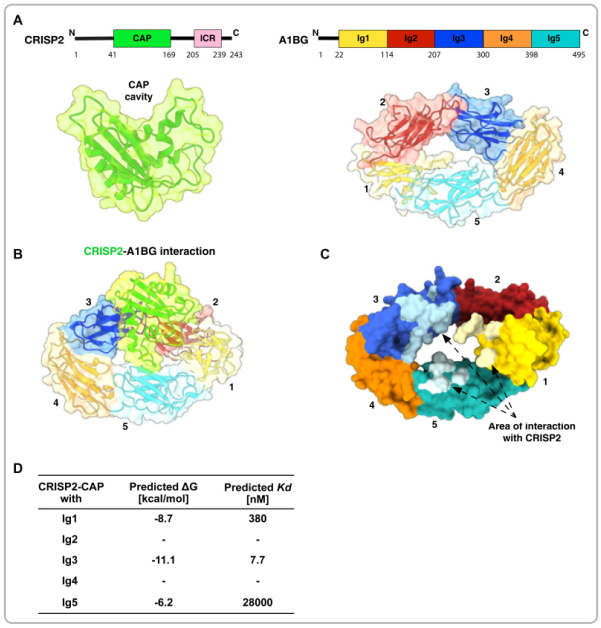
A1BG and Abacavir Interactions in *Gorilla gorilla gorilla*

A1BG overview. Alpha-1-B glycoprotein (A1BG) is a ~54 kDa plasma glycoprotein of unknown function [1] [2]. It contains five immunoglobulin-like domains (resembling antibody variable regions) [2] and is most highly expressed in liver [3]. In humans A1BG binds the cysteine-rich secretory protein CRISP3 with nanomolar affinity [4] and can inhibit sterol transport by CRISP proteins [2] [4]. A1BG has been reported as an autoantigen in rheumatoid arthritis and is upregulated in certain cancers (e.g. pancreatic adenocarcinoma) [2] [1]. Its glycosylation and repeated Ig domains suggest roles in molecular recognition, but no ligand besides CRISP3 has been identified.



CRISP2-CAP with	Predicted ΔG [kcal/mol]	Predicted *Kd* [nM]
Ig1	-8.7	380
Ig2	-	-
Ig3	-11.1	7.7
Ig4	-	-
Ig5	-6.2	28000

Figure: Predicted 3D structure of human A1BG (five Ig-like domains, colored) docking with a CAP-domain partner (green). The Ig3 domain (blue) contributes the largest interface [5] [2]. This illustrates A1BG's multi-domain fold relevant to any small-molecule binding model.

1. Gene Expression of A1BG with Abacavir

No published studies show A1BG transcription changing in response to Abacavir. In human tissue, A1BG is constitutively expressed (highest in liver [3]) and is **not** reported among genes differentially expressed after Abacavir treatment. For example, a human microarray study of subcutaneous fat in HIV patients treated with Abacavir did not list A1BG among the changed genes [6]. We found no cell- or animal-based transcriptomic data on Abacavir affecting A1BG expression. In gorillas, A1BG has an ortholog (Gogo-A1BG, gene ENSGGOG00000043585 [7]) but no expression data are available. In summary, there is **no evidence** that Abacavir induces or suppresses A1BG expression in any species.

2. Predicted A1BG–Abacavir Binding Interactions

Abacavir (a small nucleoside analog, 286 Da) is known to bind *HLA-B*57:01*, not A1BG* [8] [9] *. Structurally, unmodified Abacavir sits in the peptide-binding groove of HLA-B*57:01 (with high specificity) and alters the repertoire of presented peptides [8]. By contrast, A1BG has no known binding pocket for small molecules: its AlphaFold-predicted structure consists of five Ig-like domains in a ring (Fig. above) [5]. In silico docking of A1BG with CRISP2 (another CAP-family protein) confirmed that the Ig3 domain makes the strongest contact [5], but no docking of Abacavir to A1BG has been reported. Approximately 50% of Abacavir is protein-bound in plasma (independent of concentration) [10], but this likely reflects binding to abundant carriers (albumin and α1-acid glycoprotein) rather than to A1BG. To date, no experiments or modeling studies have identified A1BG as an Abacavir binder. Thus, any direct A1BG–Abacavir interaction remains highly speculative.

3. Pharmacogenomics (A1BG and Abacavir)

In humans, Abacavir hypersensitivity is strongly linked to the HLA-B*57:01 allele* [9] *. Clinical guidelines universally recommend screening for HLA-B*57:01 before prescribing Abacavir [9]. There is **no** analogous pharmacogenomic connection for A1BG. No A1BG gene variants are known to influence Abacavir response or toxicity, nor is A1BG included in drug–gene interaction databases. In gorillas, the equivalent of human HLA-B is *Gogo-B*; gorillas possess MHC class I alleles (Gogo-A, -B, -C) but with different specificities [11]. Gorilla A1BG (Gogo-A1BG) is conserved (orthologous to human A1BG [7]), but there are no data on gorilla Abacavir pharmacogenetics. Overall, pharmacogenetic effects on Abacavir are driven by HLA-B, not by A1BG, in humans or (presumably) in gorillas.

4. Functional Implications for Metabolism, Transport, and Immunity

A1BG is *not* an enzyme or transporter, so it likely does not affect Abacavir's metabolism or clearance. Abacavir is metabolized by hepatic alcohol dehydrogenase and UGT to inactive derivatives [12] and excreted renally. A1BG plays no known role in these pathways. In drug transport terms, Abacavir is moderately protein-bound (~50%) [10], but again A1BG has not been reported as a binding protein for Abacavir or related drugs.

Immunologically, Abacavir hypersensitivity is mediated by drug-modified self-peptide presentation by HLA-B*57:01 [8] [9]. A1BG's immune role appears limited to binding CRISP proteins, potentially modulating innate sterol transport [2] [4]. There is no evidence that A1BG presents antigens or influences T-cell responses to Abacavir. In theory, A1BG's Ig-like domains could bind peptides or small ligands, but no link to Abacavir-specific immunity has been identified.

Summary: There is currently no data supporting a direct interaction between A1BG and Abacavir in gorillas or humans. A1BG is an abundant serum protein with immunoglobulin-like domains [2], but Abacavir's known interactions (with HLA-B) and metabolism do not involve A1BG [8] [10]. In the absence of gorilla-specific studies, human/primate analogies suggest A1BG–Abacavir interactions are unlikely to be functionally significant.

Sources: Primary literature and databases as cited above; no gorilla-specific publications were found, so human and primate data have been extrapolated with caution [7] [11] [9] [8].

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